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CONCRETE MASONRY NOTES

GENERAL		
1.	MASONRY CONSTRUCTION IS TO CONFORM TO AS3700-2018	
2.	THE CHARACTERISTIC UNCONFINED COMPRESSIVE STRENGTH OF THE MASONRY UNITS SHALL BE 15MPa or GREATER.	
3.	BED UNITS IN FRESHLY PREPARED MORTAR, UNIFORMLY MIXED CEMENT, LIME AND SAND IN THE RATIO OF 1:1.6 or 10:5 CONFORMING TO AS3700-2001.	
4.	MORTAR CLASSIFICATION	M3
	DURABILITY CLASSIFICATION OF BUILT IN COMPONENTS	R1
	DURABILITY GRADE OF EXTERNAL MASONRY UNITS	PROTECTED
5.	PROVIDE TEMPORARY BRACING TO DURING CONSTRUCTION IN ACCORDANCE WITH AS3700	
6.	ALL WALLS SHOULD BE FULLY BONDED AND TIED AT CORNERS AND INTERSECTIONS UNLESS DETAILED OTHERWISE.	

LOADBEARING CONCRETE MASONRY WALLS

1. UNLESS NOTED OTHERWISE ALL CORES TO BE FILLED WITH **N20 CONCRETE GROUT** WITH A GP OR GB CEMENT RATIO OF 300kg/m³ IN ACCORDANCE CONCRETE SPECIFICATIONS.
2. CORE GROUT IS TO BE MECHANICALLY COMPACTED TO ENSURE COMPLETE FILLING OF ALL CORES.
3. CLEAN OUT BLOCKS OR E BLOCKS ARE TO BE PLACED AT THE BASE OF ALL WALLS. FULLY BED FACE SHELLS AFTER BASE HAS BEEN CLEANED AND INSPECTED.
4. THE VERTICAL REINFORCEMENT TO STARTER BARS AT BASE OF WALLS. STARTER BARS TO BE SAME SIZE AND SPACING AS VERTICAL REINFORCEMENT
5. UNLESS NOTED OTHERWISE IN DOCUMENTATION LOAD BEARING WALLS TO BE REINFORCED WITH:
190 BLOCK – N16-400 VERTICAL, N12-400 HORIZONTAL BARS LOCATED CENTRALLY.
290 BLOCK – N16-400 EACH FACE VERTICAL 50 COVER, N16-400 HORIZONTAL CENTRAL.
6. LAP REINFORCEMENT 40 BAR DIAMETERS UNLESS NOTED OTHERWISE.
7. PROVIDE DOWNED EXPANSION JOINTS AT 12 METER MAXIMUM CENTERS AND WHERE WALLS CROSS SLAB JOINTS. EXPANSION JOINTS TO BE LOCATED MINIMUM 400 AWAY FROM DOORS AND WINDOWS.
8. WALLS UNDER AND SUPPORTING CONCRETE SLABS TO BE FINISHED WITH STEEL TROWEL FLUSH WITH SOFFIT OF SLAB OR BEAM OVER. AT TERMINATING WALLS REINFORCEMENT TO BE COGGED INTO SLAB. FOR WALLS CONTINUING OVER SLAB, REINFORCEMENT TO BE LAPPED THROUGH SLAB DEPTH.
9. WALLS UNDER POST TENSIONED REINFORCED SLABS TO BE FINISHED WITH A SLIP JOINT COMPRISING OF TWO LAYERS OF GREASED ANNEALED ALUMINUM SHEETS AND 50 DIA CONDUIT OVER VERTICAL REINFORCEMENT.
10. BLOCK LINTELS TO BE PROPPED UNTIL WALL OVER IS POURED AND CURED.

NON LOADBEARING CONCRETE MASONRY WALLS

1. ALL EXTERNAL NON LOAD BEARING WALLS TO BE CORE FILLED AND REINFORCED IN ACCORDANCE WITH LOAD BEARING CONCRETE MASONRY WALLS.
2. WALLS TO BE FINISHED WITH STEEL TROWEL FLAT AND EVEN 20 BELOW SOFFIT OF STRUCTURE OVER. FILL GAP WITH BACKING ROD AND SEALANT REFER ARCHITECT FOR FIRE RATING REQUIREMENTS.
3. TOP OF NON LOAD BEARING WALLS TO BE RESTRAINED BY EITHER CONNECTING TO CONCRETE SOFFITS WITH MET-4 SLIDING TIE AT 400 CENTERS OR TO STEELWORK WITH APPROPRIATE SLIDING MASONRY TIE AT 400 CENTERS SHOT FIXED TO STEEL.
4. WALLS ADJACENT STEEL COLUMNS ARE TO BE SHOT FIXED TO COLUMNS WITH SLIDING MASONRY TIE AT 400 MAX CENTERS.
5. FOR NON CORE FILLED WALLS GROUT CORES ADJACENT DOORWAYS, WINDOWS, EXPANSION JOINTS AND AT 3.0m MAX CENTERS. PROVIDE #16 REINFORCEMENT BAR CENTRAL. DRILL 125 AND EPOXY STARTER BARS TO SLAB.
6. NON LOAD BEARING WALLS TO BE CONSTRUCTED ON 150 MINIMUM THICK SLAB. IF SLAB THICKNESS IS LESS THAN 150 PROVIDE 600 wide 200 DEEP THICKENING UNDER WALL WITH S181 MESH BOTTOM.
7. UNLESS NOTED OTHERWISE ALL NON LOADING BLOCK WALL OPENINGS ARE TO HAVE A PERMANENT GALVANIZED STEEL ANGLE LINTEL. OPENINGS UP TO 2000 USE 90x8 EQUAL ANGLE. FOR OPENINGS UP TO 3000 USE 150x90x10 UNEQUAL ANGLE WITH THE LONG SIDE VERTICAL. ALL LINTELS TO HAVE 200 MINIMUM BEARING EACH END.

CONCRETE MASONRY CANTILEVER RETAINING WALLS

1. UNLESS NOTED OTHERWISE ALL CORES TO BE FILLED WITH N20 CONCRETE GROUT WITH A GP OR GP CEMENT RATIO OF 300kg/m³ IN ACCORDANCE CONCRETE SPECIFICATIONS.
2. PROVIDE 100 DIA SLOTTED PVC PIPE IN GEOTEXTILE SOCK TO EARTH SIDE OF RETAINING WALLS WITH ADEQUATE FALLS TO OUTLET.
3. DIRECTLY BEHIND RETAINING WALL BACKFILL 300 MIN WIDE SECTION WITH FREE DRAINING GRANULAR AGGREGATE
4. ENSURE BACKFILL COMPACTION DOES NOT EXCEED SURCHARGE LOADING OF RETAINING WALL.
5. BACK PROP RETAINING WALLS DURING COMPACTION UNTIL CORE FILL HAS CURED FOR 7 DAYS MINIMUM.
6. CLEAN OUT BLOCKS OR E BLOCKS ARE TO BE PLACED AT THE BASE OF ALL WALLS. FULLY BED FACE SHELLS AFTER BASE HAS BEEN CLEANED AND INSPECTED.

INDUSTRIAL PAVEMENT SLAB ON GROUND

SUBGRADE	1. ANY MOVEMENT, SWELLING, HEAVING, OR SHRINKAGE TO THE SUBGRADE WILL HAVE A DIRECT IMPACT ON THE PERFORMANCE OF THE SLAB ON GROUND CONCRETE ABOVE. MINOR SOFT SPOTS IN THE SUBGRADE CAN CAUSE CRACKING AND DEFORMATION TO THE CONCRETE. 2. A PROOF ROLL IS REQUIRED TO BE CARRIED OUT BY THE CONTRACTOR, UNDER GEOTECHNICAL SUPERVISION, TO ALL SUBGRADES SUPPORTING STRUCTURAL SLABS ON GROUND. ALL SOFT SPOTS ARE TO BE IDENTIFIED AND RECTIFIED IN ACCORDANCE WITH GEOTECHNICAL ENGINEERING INSTRUCTIONS. 3. FOR MEDIUM AND LARGE SCALE BULK EARTHWORK OPERATIONS (CUT AND FILL), A NEW GEOTECHNICAL INVESTIGATION MAY BE REQUIRED TO RETEST THE NEW STRENGTH AND SERVICEABILITY CHARACTERISTICS OF THE SLAB SUBGRADE. 4. THE RACKING SUPPLIER AND BUILDINGS OPERATIONAL TEAM NEED TO REVIEW THE EXPECTED SITE SETTLEMENTS FOR POTENTIAL SERVICEABILITY CONSTRAINTS TO THE BUILDINGS RACKING AND EQUIPMENT. 5. PLATE LOAD TESTING (PLT) SHOULD BE UNDERTAKEN ON THE SUBGRADE PRIOR TO THE PLACING OF SUBBASE. THE PLT INVESTIGATION ON SITE SHOULD BE CARRIED OUT IN ACCORDANCE WITH BS EN 1997-2 EUROCODE 7. MINIMUM REQUIRED MODULUS OF SUBGRADE REACTION OF 35kN/m²/mm IS REQUIRED TO MEET DESIGN ASSUMPTIONS. ANY AREAS NOT MEETING THIS REQUIREMENT ARE TO BE REMEDIATED AND RECOMPACTED BASED ON GEOTECHNICAL ADVICE. SEEK APPROVAL FROM THE DESIGN ENGINEER BEFORE PLACING SUBBASE. GENERALLY, THERE SHOULD BE A MINIMUM OF ONE PLATE LOADING TEST PER 200m² OF FLOOR.
SUBBASE	1. A 150mm CBR15 COMPRESSED MATERIAL SUBBASE IS HIGHLY RECOMMENDED TO BE INSTALLED UNDERNEATH ALL STRUCTURAL SLABS ON GROUND AS IT IMPROVES JOINT LOAD TRANSFER, PROVIDES A FLAT AND STABLE WORKING PLATFORM, FACILITATES A UNIFORM BEARING SURFACE, AND PREVENTS PUMPING AT THE JOINTS AND PAVEMENT EDGES. 2. THE SUBBASE SHALL BE LEVELLED TO A TOLERANCE OF +0 / -15mm, WITH A SURVEY PERFORMED AND SUBMITTED TO NORTHROP CONSULTING ENGINEERS PRIOR TO LAYING OF THE MEMBRANE.

PRECAST NOTES

GENERAL

1. ALL PRECAST CONCRETE WORK SHALL BE CARRIED OUT IN ACCORDANCE WITH AS3850:1, AS3850:2, AS3600 AND AS3610:1 AS APPROPRIATE.
2. THE PRECAST CONCRETE SUPPLIER SHALL BE RESPONSIBLE FOR THE DETAILING OF PRECAST ELEMENTS AND ANY CONNECTIONS NOT SHOWN ON THE STRUCTURAL DRAWINGS.
3. REINFORCEMENT SHOWN ON STRUCTURAL DRAWINGS IS THE MINIMUM REQUIRED FOR IN-SERVICE CONDITIONS. THE SUPPLIER IS RESPONSIBLE FOR ANY ADDITIONAL REINFORCEMENT REQUIRED FOR LOADS DUE TO HANDLING, PROPPING, TRANSPORTATION AND ERECTION. A SUITABLY QUALIFIED ENGINEER SHALL BE ENGAGED (BY OTHERS) TO FILL THE ROLE OF "ERECTION DESIGN ENGINEER". THE ERECTION DESIGN ENGINEER SHALL BE RESPONSIBLE FOR THE DESIGN OF ALL ELEMENTS FOR CASTING, HANDLING, TRANSPORTATION & ERECTION. THEY SHALL PROVIDE WRITTEN CERTIFICATION TO NORTHPRO CONSULTING ENGINEERS PRIOR TO FABRICATION.
4. THE STRUCTURAL DRAWINGS HAVE BEEN PREPARED TO INDICATE THE STRUCTURAL INTENT. THE SUPPLIER SHALL PREPARE SHOP DRAWINGS SHOWING DIMENSIONS OF ALL PANELS INCLUDING CHAMFERS, DRIP GROOVES AND REBATES FOR WINDOWS, ETC, AND ALL STRUCTURAL DETAILS. WHERE NECESSARY THE SHOP DETAILER IS TO MAKE ASSUMPTIONS AND SUBMIT TO NORTHPRO CONSULTING ENGINEERS FOR RESOLUTION. SHOP DETAILER IS TO ALLOW TO RE-WORK SHOP DRAWINGS AS REQUIRED, THESE SHALL BE SUBMITTED, ALONG WITH CALCULATIONS, TO THE BUILDER FOR THEIR APPROVAL. BUILDER SHALL LODGE TWO HARD COPIES OF APPROVED DRAWINGS TO NORTHPRO CONSULTING ENGINEERS FOR REVIEW **ALLOW 5 WORKING DAYS FOR REVIEW**. THESE DRAWINGS SHALL COVER THE HANDLING PROCEDURE OF THE UNITS THROUGHOUT ALL STAGES INCLUDING STRIPPING, LIFTING, STACKING, TRANSPORTING AND ERECTION. CONCRETE STRESSES THROUGHOUT HANDLING SHALL NOT CAUSE CRACKING. COMPUTATIONS AND DETAILS SHALL INCLUDE LOCATION AND SIZE OF INSERTS AND TESTS PROVING ANCHORAGE CAPACITY OF LIFTING FERRULES.
5. COLOUR AND FINISH SAMPLES OF PRECAST ELEMENTS SHALL BE SUBMITTED TO THE ARCHITECT FOR APPROVAL BEFORE ELEMENTS ARE CAST.
6. THE CONTRACTOR SHALL ENSURE THAT CRANE OUTRIGGER LOADS ARE SPREAD AND LOCATED TO AVOID CRACKING OF THE GROUND SLAB AND UNDER NO CIRCUMSTANCES LOAD SLAB WITHIN 500mm of a FREE EDGE.
7. THE SUPPLIER SHALL ENSURE THE BUILDER IS GIVEN ADEQUATE NOTICE OF FIXING DETAILS WHICH MAY AFFECT THE CONSTRUCTION OF THE STRUCTURE, E.G. INSERTS IN BEAMS, HOLES IN STRUCTURAL STEEL MEMBERS, ETC.
8. CRANES WILL NOT BE PERMITTED ON ADJACENT PROPERTY FOR ERECTION OF TILT PANELS.
9. PRECAST PANEL CONCRETE TO BE 'S' CLASS IN ACCORDANCE WITH CONCRETE SPECIFICATIONS
10. THE CONCRETE STRENGTH AT REMOVAL FROM MOULDS IS TO BE A MINIMUM OF **20MPa** UNO
11. THE PANELS MUST BE STACKED, STORED AND TRANSPORTED IN SUCH A WAY THAT, CRACKING WILL NOT OCCUR, AND OR WARPING IN EXCESS OF THAT GIVEN IN THE RELEVANT CODES WILL NOT OCCUR.
12. LIFTING FIXINGS - THE CONTRACTOR SHALL SUPPLY AND FIT LIFTING FIXINGS AS REQUIRED. THESE SHALL TAKE THE FORM OF CAST IN CABLES OR FERRULES. THEY SHALL NOT BE LOCATED IN THE FACE WHICH IS EXPOSED TO VIEW IN THE FINAL CONDITION AND AFTER USE SHALL BE PROTECTED TO AVOID CORROSION AND STAINING.
13. STEEL CONNECTIONS TO PRECAST PANELS TO BE FULLY GROUTED WITH NON SHRINK 40MPa GROUT PLACED IN SUCH A WAY AS TO NOT SPOIL THE AESTHETICS OF THE PANEL AND NOT FILL OR LOCK PANEL JOINTS.
14. ALL TEMPORARY BRACES SHALL BE FIXED TO PANELS AND BRACING SUPPORTS WITH MECHANICAL INSERTS SUCH AS CAST-IN, DRILLED THROUGH OR UNDERCUT ANCHORS. UNO MASONRY EXPANSION ANCHORS MAY ONLY BE USED WITH THE PRIOR WRITTEN APPROVAL OF NORTHPRO CONSULTING ENGINEERS. CHEMICAL ANCHORS MAY ONLY BE USED IF EACH ANCHOR IS PROOF LOAD TESTED TO ITS WORKING LOAD LIMIT AND WRITTEN PERMISSION HAS BEEN OBTAINED FROM NORTHPRO CONSULTING ENGINEERS.
15. ALL FIXINGS IN FIRE RATED WALLS AND FIRE RATED EXTERNAL WALLS TO COMPLY WITH SPECIFICATION C111 OF THE BCA.
16. THE CONTRACTOR SHALL ENSURE THAT THE PANEL STACKING HEIGHTS AND LOCATIONS DO NOT EXCEED THE GROUND SLAB CAPACITY AND SHALL UNDER NO CIRCUMSTANCES STACK PANELS MORE THAN 5 HIGH.
17. THE CONTRACTOR SHALL ENGAGE A SUITABLY QUALIFIED STRUCTURAL ENGINEER TO INSPECT THE REINFORCEMENT OF ALL ELEMENTS PRIOR TO CONCRETE PLACEMENT. THEY ARE TO PROVIDE A WRITTEN CERTIFICATION TO NORTHPRO ENGINEERS STATING THAT ELEMENTS HAVE BEEN CONSTRUCTED IN ACCORDANCE WITH NORTHPRO DRAWINGS AND DRAWINGS FROM "ERECTION DESIGN ENGINEER".

ABBREVIATIONS

GENERAL	ABR	CONCRETE	ABR	STEELWORK GENERAL CONT.	ABR
AUSTRALIAN HEIGHT DATA	AHD	CONSTRUCTION JOINT	CJ	CONTINUOUS FILLET WELD	CFW
BOTTOM	BTM	DOWEL JOINT	DJ#	COMP. PENETRATION BUT WELD	CPBW
CENTRE LINE	CL	EDGE BEAM	EB#	SQUARE HOLLOW SECTION	SHS
CENTERS	CTS	EDGE THICKENING	ETH	RECTANGLE HOLLOW SECTION	RHS
CONCRETE	CONC	INTERNAL BEAM	IB#	PLATE	PL
CONCRETE COLUMN MARK	CC#	ISOLATION JOINT	IJ#	GENERAL PURPOSE BRACKET	GPB
DEEP/ DEPTH	D	PERMANENT MOVEMENT JOINT	MJ	OVER	/
DIAMETER	DIA	POST TENSIONED	PT	PARALLEL FLANGE CHANNEL	PFC
DRAWING	DRG	SAWN JOINT	SJ#	STEELWORK TAGS	ABR
EQUAL	EQ	SLAB THICKENING	TH#	BARGE PURLIN	BP#
EXISTING	EX	FOOTINGS		BEAM/ BEARER	B#
FINISHED FLOOR LEVEL	FFL	BORED PIER	BP#	LINTEL	L#
HORIZONTAL	H	FOOTING BEAM	FB#	FASCIA PURLIN	FP#
IMPOSED ACTION-LIVE LOAD	Q	STRIP FOOTING	SF#	JOIST	J#
INTERSECTION POINT	IP	PAD FOOTING	PF#	FLY BRACE	FB#
INVERT LEVEL	IL	PILE	P#	GIRDER TRUSS	GT#
		BORE HOLE	BH#	GIRT	G#
MAXIMUM	MAX	TEST PIT	TP#	HANGING COLUMN	HC#
METER	m	REINFORCEMENT		HIP TRUSS	HT#
MINIMUM	MIN	BAR LAYING SEQUENCE	1st	JACK TRUSS	JT#
MISCELLANEOUS	MISC	BOTTOM REINFORCEMENT	B	MULLION	M#
NOMINAL	NOM	PLACED CENTRAL C	C	PARALLEL GIRDER TRUSS	PGT#
NOT SHOWN ON PLAN	NSOP	EXTRA REINF.	E	PURLIN	P#
NOT SHOWN ON ELEVATION	NSOE	PLACED EACH FACE	EF	RAFTER (RAKING BEAM)	R#
NOT TO SCALE	NTS	PLACED EACH WAY	EW	RAKING ANGLE	RA#
OPPOSITE	OPP	PLACED FAR FACE	FF	ROOF BRACING	RB#
OVERALL	O/A	PLACED HORIZONTAL	H	STEEL COLUMN	SC#
PAD LEVEL	PL	PLACED NEAR FACE	NF	STAIR STRINGER	ST#
PERMANENT ACTION-DEAD LOAD	G	TOP REINFORCEMENT	T	STRUT	S#
QUANTITY	QTY	PLACED VERTICAL	V	TRUNCATED GIRDER TRUSS	TGT#
RADIUS	R	STEELWORK GENERAL		TRUSS	T#
REDUCED LEVEL	RL	CIRCULAR HOLLOW SECTION	CHS	VALLEY TRUSS	VT#
REINFORCEMENT	REINF	EQUAL ANGLE	EA	WALL ANGLE	WA#
SQUARE	SQ	FLAT	FL	WALL BRACE	WB#
TOP OF FOOTING	TOF	GALVANISED	GALV	WALL HEAD	WH#
TOP OF WALL	TOW	UNDER	/U	WALLS	
TYPICAL	TYP	UNEQUAL ANGLE	UA	BLOCK WALL	BW#
UNDERSIDE	U/S	UNIVERSAL BEAM	UB	BRICK WALL	BK#
UNLESS NOTED OTHERWISE	UNO	UNIVERSAL COLUMN	UC	CONCRETE WALL	CW#
VERTICAL	V	SNUG TIGHT	/S	NOT SHOWN ON ELEVATION	NSOE
WIDE / WIDTH	w	TENSION AND BEARING TIGHT	/TB	PRECAST WALL	PW#
				RETAINING WALL	RW#

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						Design Coordinated: HP	Compliance Checked: --		
						Authorised For Issue: NS	Construction Checked: --		
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Drawing Number	Project No	Originator	Volume	Level	Type	Discipline	Number
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